

The `coppe` document class

Version 3.5.1

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BIB_TE_X and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

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copying, distributing or modifying the source code, among other acts covered by this license.

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3 Support

We maintain a mailing list where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain
`\mainmatter` three-parts: *front*, *main* and *back* matters [?]. Each one of these parts is started
`\backmatter` by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`.
The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Front cover

This element was recently introduced by the Academic Registry. It is automatically constructed by the `\maketitle` command.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation [?]. It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your
`\foreigntitle` monograph in the current and foreign languages. The default languages are Brazilian Portuguese and English. The `babel` package is automatically loaded by `coppe.cls`, so you do not need to load it again. The Brazilian Portuguese is the main language and the English is only required for the foreign abstract.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc.

`\examiner` students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

`\advisor{Title}{Advisor's Name}{Surname}{Degree}`

```
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract` (*env.*) As stated by the Academic Registry [?], abstracts must be in one page each, with `foreignabstract` (*env.*) at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.

`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is provided, it will be used as sort key. This was later, in 2024, implemented also for `\printloabbreviations` and `\printlosymbols`. `\abbrev`, otherwise `Symbol`, or `Abreviation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

COPPE_T_E_X produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no BIB_T_E_X `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` BIB_T_E_X styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib} % in the preamble
...
\printbibliography           % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard `LATEX` command used before all appendices. `COPPETEX` introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for `backmatter` is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsceexam, msceexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msceexam`, and `dsceexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

english `CoppeTEX` uses `Babel`. The default language is Portuguese (actually `brazilian`), with English being the second language. If option `english` is used, English becomes the main language and Portuguese the secondary. Look at the `Babel` package to switch between languages.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

6 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and `pdflatex` will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package. It is also useful to enable hyperlinks. Fortunately, the `coppe` class will do this automatically if `hyperref` is loaded.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

`longquote` **Quotation** To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

7 A simple example

```
1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{booktabs}% tabelas mais bonitas
5 \usepackage{rotating}% rodando coisas, como tabelas
6 \usepackage{longtable}% tabelas longas
```

```

7 \usepackage[most]{tcolorbox} % caixas de texto
8 \usepackage{amsmath,amssymb}
9 \usepackage{hyperref}
10 % listings, fancyvrb e algorithm2e já vêm carregados pela classe coppe (v3.8).
11
12 \addbibresource{example.bib}% base de referências (biblatex/biber)
13
14 \makelossymbols
15 \makeloabbreviations
16
17 \begin{document}
18 \title{Título da Tese}
19 \foreigntitle{Thesis Title}
20 \author{Nome do Autor}{Sobrenome}
21 \advisor{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
22 \advisor{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
23 \advisor{Prof.}{Nome do Terceiro Orientador}{Sobrenome}{D.Sc.}
24
25 \examiner{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
26 \examiner{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
27 \examiner{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
28 \examiner{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
29 \examiner{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
30 \department{PESC}
31 \date{01}{2024}
32
33 \keyword{Primeira palavra-chave}
34 \keyword{Segunda palavra-chave}
35 \keyword{Terceira palavra-chave}
36
37 \maketitle
38
39 \frontmatter
40 \dedication{A alguém cujo valor é digno desta dedicatória.}
41
42 \chapter*{Agradecimentos}
43
44 Gostaria de agradecer a todos os que contribu\iram para este trabalho:
45 orientadores, colegas, familiares e \as ag\ências de fomento. Lorem ipsum
46 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
47 ut labore et dolore magna aliqua.
48
49 \begin{abstract}
50
51 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
52 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
53 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
54 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
55 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
56 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
57 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
58 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
59 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
60 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit

```

```

61 aspernatur aut odit aut fugit.
62
63 \end{abstract}
64
65 \begin{foreignabstract}
66
67 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
68 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
69 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
70 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
71 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
72 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
73 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
74 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
75 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
76 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
77 aspernatur aut odit aut fugit.
78
79 \end{foreignabstract}
80
81 \tableofcontents
82 \listoffigures
83 \listoftables
84 \listofquadros
85 \listofprogramas
86 \listofalgorithms
87 \printlosymbols
88 \printloabbreviations
89
90 \mainmatter
91 \chapter{Introdução}
92
93 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da
94
95 A classe \verb|coppe| foi criada por Vicente Helano e George Ainsworth, porém, em 2024, é
96
97 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.com}
98
99 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
100
101 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
102
103 É importante de notar que essa classe não foi construída sobre a classe \LaTeX \ para a A
104
105 Apesar desse modelo ser muito bom, ele tem um defeito: a limitação do sistema de referênci
106
107 Mais ainda, as regras da COPPE ainda não se adaptaram, no início de 2024, as novas regras
108
109 Este documento não substitui, mas complementa, o documento que descreve a classe.
110
111
112
113 \chapter{Configurações Iniciais}
114

```


115 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c
116
117 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
118
119 Recomendo ler o documento “The \verb*|coppe| document class” para entender melhor todas
120
121 \section{Linguagem principal do texto}
122
123 Essa classe considera que o texto principal está em português e algumas partes específicas
124 \begin{itemize}
125 \item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.
126 \item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
127 \end{itemize}
128
129 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe Coppe\T
130
131
132 \section{Por que usar o \LaTeX}
133
134 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso dess
135
136 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se p
137
138 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob
139
140 \section{Como e onde usar o \LaTeX}
141
142 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambiente
143 \begin{enumerate}
144 \item Na sua máquina, instalando uma versão completa como o MikTeX, típico do Windows, ou
145 \item Usar um ambiente na rede, como o Overleaf.
146 \end{enumerate}
147
148 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b
149
150
151 \chapter{Algumas Regras da COPPE}
152
153 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei
154
155 É imprescindível definir os símbolos, tal como o
156 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
157 \syml{\mathbb{R}}{Conjunto dos números reais}
158 \syml{\emptyset}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como
159
160 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \ve
161
162 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice
163
164 \section{Citações}
165
166 Citações curtas podem ser feitas \quote{o comando quote} ou direto com “duas crases e dois
167
168 \begin{longquote}

```

169 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
170 feita com o ambiente \verb=longquote= The primary objective of this
171 investigation was to determine the feasibility of detecting corrosion in
172 aluminum Naval aircraft components with neutron radiographic interrogation
173 and the use of standard corrosion penetrameters. Secondary objectives
174 included the determination of the effect of object thickness on image quality,
175 the defining of minimum levels of detectability and a preliminary investigation
176 of a means whereby the degree of corrosion could be quantified with neutron
177 radiographic data. \cite{article-example}
178 \end{longquote}
179
180 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \verb*|na
181
182 Em todo caso, \textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em
183
184 \chapter{Floats}
185
186 Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \texti
187
188 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm
189
190 \textbf{Pela norma atual (manual2024), a legenda} \verb|\caption| \textbf{de toda ilustração}
191
192 Quadros são ilustrações cujos dados são delimitados por linhas em \emph{todas} as bordas (ao
193
194
195 \section{Tabelas e Figuras Padrão}
196
197 Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.
198
199 \begin{table}[ht]
200 \centering % Centraliza a tabela
201 \caption{Exemplo de Tabela de Números}
202 \label{tab:exemplo_numeros}
203 \begin{tabular}{ccc} % Define a quantidade de colunas
204 \hline % Linha superior
205 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ % Cabeçalhos
206 \hline % Linha média
207 1 & 2 & 3 \\ % Primeira linha de dados
208 \hline
209 4 & 5 & 6 \\ % Segunda linha de dados
210 \hline
211 7 & 8 & 9 \\ % Terceira linha de dados
212 \hline
213 10 & 11 & 12 \\ % Quarta linha de dados
214 \hline % Linha inferior
215 \end{tabular}
216 \fonte{Elabora\c{c}\~ao pr'opria.} % fonte obrigatoria, abaixo do float
217 \end{table}
218
219
220
221 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
222

```

```

223 \begin{figure}[ht]
224 \centering % Centraliza a figura
225 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
226 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
227 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
228 \source{COPPE/UFRJ.} % fonte obrigatória, abaixo da figura
229 \end{figure}
230
231 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
232
233 \begin{quadro}[ht]
234 \centering
235 \caption{Exemplo de Quadro}
236 \label{qua:exemplo}
237 \begin{tabular}{|l|l|}
238 \hline
239 \textbf{Ilustração} & \textbf{Delimitação} \\
240 \hline
241 Tabela & Linhas apenas no topo e na base. \\
242 \hline
243 Quadro & Linhas em todas as bordas. \\
244 \hline
245 \end{tabular}
246 \fonte{Elabora\c{c}\~ao pr\'opria.}
247 \end{quadro}
248
249
250
251
252
253 \section{Tabelas mais elegantes}
254
255 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
256
257 \begin{table}[ht]
258 \centering % Centraliza a tabela
259 \caption{Exemplo de Tabela de Números mais elegantes}
260 \label{tab:exemplo_numerosbom}
261 \begin{tabular}{ccc} % Define a quantidade de colunas
262 \toprule % Linha superior
263 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\
264 \midrule % Linha média
265 1 & 2 & 3 \\
266 4 & 5 & 6 \\
267 7 & 8 & 9 \\
268 10 & 11 & 12 \\
269 \bottomrule % Linha inferior
270 \end{tabular}
271 \end{table}
272
273 \section{Tabelas Longas ou Largas}
274
275 Se sua tabela é muito longa ou larga, existem várias opções.
276 \begin{itemize}

```

```

277 \item alterar o tamanho da letra
278 \item Usar o longtable
279 \item rodar a tabela, fazendo ela em \textit{landscape}
280 \item fazer a tabela dentro de um minibox
281 \end{itemize}
282
283
284 \subsection{Tabelas largas demais}
285
286 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
287
288 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
289
290 \begin{table}[ht]
291 \centering % Centraliza a tabela
292 \caption{Exemplo de Tabela Larga com Fonte Menor}
293 \label{tab:tabela_largafns}
294 \footnotesize % Aplica uma fonte menor para a tabela
295 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
296 \toprule
297 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} & \textbf{Coluna 4} & \textbf{Coluna 5} & \textbf{Coluna 6} & \textbf{Coluna 7} \\
298 \midrule
299 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
300 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
301 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
302 \bottomrule
303 \end{tabular}
304 \end{table}
305
306 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
307
308 \begin{table}[ht]
309 \centering
310 \caption{Exemplo de Tabela Redimensionada}
311 \label{tab:examplerb}
312 \resizebox{\textwidth}{!}{%
313 \begin{tabular}{llll}
314 \toprule
315 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
316 \midrule
317 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
318 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
319 \bottomrule
320 \end{tabular}%
321 }
322 \end{table}
323
324
325 Para rodar uma tabela muito larga em 90 graus no LaTeX, você pode usar o pacote \verb*|rotat
326
327 Aqui está um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
328
329 \begin{sidewaystable}
330 \centering

```

```

331 \caption{Sua Legenda Aqui}
332 \label{tab:sua_tabela}
333 \begin{tabular}{lll}
334 \toprule
335 Coluna 1 & Coluna 2 & Coluna 3 \\
336 \midrule
337 Item 1 & Item 2 & Item 3 \\
338 Item 4 & Item 5 & Item 6 \\
339 \bottomrule
340 \end{tabular}
341 \end{sidewaystable}
342
343 Se a tabela for muito longa, o ambiente \verb|longtable| é o ideal. Ele fornece comandos par
344
345 % Exemplo de tabela longa que se estende por várias páginas
346 \begin{longtable}{|c|c|c|}
347 % primeiro cabeçalho (é o caption)
348 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
349 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
350 \endfirsthead
351 % cabeçalho normal
352 \multicolumn{3}{c}%
353 {\table\ -- continuação da página anterior} \\
354 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
355 \endhead
356 % pé normal
357 \hline \multicolumn{3}{r|}{Continua na próxima página} \\ \hline
358 \endfoot
359 \hline
360 % último pé
361 \multicolumn{3}{r|}{Continua na próxima página} \\
362 \hline \hline
363 \endlastfoot
364
365 % Conteúdo da tabela
366 1 & 2 & 3 \\
367 4 & 5 & 6 \\
368 1 & 2 & 3 \\
369 4 & 5 & 6 \\
370 1 & 2 & 3 \\
371 4 & 5 & 6 \\
372 1 & 2 & 3 \\
373 4 & 5 & 6 \\
374 1 & 2 & 3 \\
375 4 & 5 & 6 \\
376 1 & 2 & 3 \\
377 4 & 5 & 6 \\
378 1 & 2 & 3 \\
379 4 & 5 & 6 \\
380 1 & 2 & 3 \\
381 4 & 5 & 6 \\
382 1 & 2 & 3 \\
383 1 & 2 & 3 \\
384 4 & 5 & 6

```

385 1 & 2 & 3 \\
386 4 & 5 & 6 \\
387 1 & 2 & 3 \\
388 4 & 5 & 6 \\
389 1 & 2 & 3 \\
390 4 & 5 & 6 \\
391 1 & 2 & 3 \\
392 4 & 5 & 6 \\
393 1 & 2 & 3 \\
394 4 & 5 & 6 \\
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396 4 & 5 & 6 \\
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398 4 & 5 & 6 \\
399 1 & 2 & 3 \\
400 4 & 5 & 6 \\
401 1 & 2 & 3 \\
402 4 & 5 & 6 \\
403 1 & 2 & 3 \\
404 4 & 5 & 6 \\
405 1 & 2 & 3 \\
406 4 & 5 & 6 \\\ 1 & 2 & 3 \\
407 4 & 5 & 6 \\
408 1 & 2 & 3 \\
409 4 & 5 & 6 \\
410 1 & 2 & 3 \\
411 1 & 2 & 3 \\
412 4 & 5 & 6 \\
413 1 & 2 & 3 \\
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437 1 & 2 & 3 \\
438 4 & 5 & 6 \\

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439 1 & 2 & 3 \\
440 4 & 5 & 6 \\
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464 4 & 5 & 6 \\
465 1 & 2 & 3 \\
466 4 & 5 & 6 \\
467 1 & 2 & 3 \\
468 4 & 5 & 6 \\
469 1 & 2 & 3 \\
470 4 & 5 & 6 \\
471 1 & 2 & 3 \\
472 4 & 5 & 6 \\
473
474 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
475 \end{longtable}
476
477 \chapter{Revis\~ao Bibliogr\'afica}
478
479 Para ilustrar a completa ades\~ao ao estilo de cita{\c c}\~oes e listagem de
480 refer\^encias bibliogr\'aficas, a Tabela~\ref{tab:citation} apresenta cita{\c
481 c}\~oes de alguns dos trabalhos contidos na norma fornecida pela CPGP da
482 COPPE, utilizando o estilo numérico. Tirando do comando inicial o parâmetro opcional numér
483
484 \begin{table}[h]
485 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
486 \texttt{\textbackslash cite} do \LaTeX\ e
487 o comando \texttt{\textbackslash citet},
488 fornecido pelo pacote \texttt{natbib}.}
489 \label{tab:citation}
490 \centering
491 {\footnotesize
492 \begin{tabular}{|c|c|c|}

```

```

493 \hline
494 Tipo da Publicação & \verb|\cite| & \verb|\citet|\
495 \hline
496 Livro & \cite{book-example} & \citet{book-example}\
497 Artigo & \cite{article-example} & \citet{article-example}\
498 Relatório & \cite{techreport-example} & \citet{techreport-example}\
499 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\
500 Anais de Congresso & \cite{inproceedings-example} &
501 \citet{inproceedings-example}\
502 Séries & \cite{incollection-example} & \citet{incollection-example}\
503 Em Livro & \cite{inbook-example} & \citet{inbook-example}\
504 Dissertação de mestrado & \cite{mastersthesis-example} &
505 \citet{mastersthesis-example}\
506 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\
507 \hline
508 \end{tabular}}
509 \end{table}
510
511 \begin{table}[h]
512 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
513 \texttt{\textbackslash cite} do \LaTeX\ e
514 o comando \texttt{\textbackslash citet},
515 fornecido pelo pacote \texttt{natbib}. Além disso, usando o booktabs.}
516 \label{tab:citation1}
517 \centering
518 {\footnotesize
519 \begin{tabular}{ccc}
520 \toprule
521 Tipo da Publicação & \verb|\cite| & \verb|\citet|\
522 \midrule
523 Livro & \cite{book-example} & \citet{book-example}\
524 Artigo & \cite{article-example} & \citet{article-example}\
525 Relatório & \cite{techreport-example} & \citet{techreport-example}\
526 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\
527 Anais de Congresso & \cite{inproceedings-example} &
528 \citet{inproceedings-example}\
529 Séries & \cite{incollection-example} & \citet{incollection-example}\
530 Em Livro & \cite{inbook-example} & \citet{inbook-example}\
531 Dissertação de mestrado & \cite{mastersthesis-example} &
532 \citet{mastersthesis-example}\
533 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\
534 \bottomrule
535 \end{tabular}}
536 \end{table}
537
538 \chapter{Alguns outros exemplo úteis}
539
540 \begin{tcolorbox}[title=Meu Textbox]
541 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
542 \end{tcolorbox}
543
544 \begin{tcolorbox}
545 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
546 \end{tcolorbox}

```



```

547
548 \begin{figure}[ht]
549     \centering
550     \begin{tikzpicture}
551         \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{image.png}}
552         \begin{scope}[x={(image.south east)},y={(image.north west)}]
553             % Definindo o textbox dentro da figura
554             \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=Este textbox fala sobre como inserir um textbox dentro de uma figura usando tikz]
555                 \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de exemplo]
556                     Este textbox fala sobre como inserir um textbox dentro de uma figura usando tikz
557                 \end{tcolorbox}
558             };
559         \end{scope}
560     \end{tikzpicture}
561     \caption{Figura com Textbox}
562     \label{fig:figura_com_textbox1}
563 \end{figure}
564
565
566 \begin{figure}[ht]
567     \centering
568     \begin{tcolorbox}
569         Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como uma imagem.
570     \end{tcolorbox}
571     \caption{Figura com Textbox simples}
572     \label{fig:figura_com_textbox}
573 \end{figure}
574
575 \section{Listagens de código e algoritmos}
576
577 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a algoritmos (pacote \verb|algorithm|).
578
579 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
580 def saudacao(nome):
581     # imprime uma saudação acentuada
582     print(f"Olá, {nome}!")
583
584 saudacao("COPPE")
585 \end{python}
586 \source{Elaboração própria.}
587
588 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|, comando \verb|listings|.
589
590 \begin{algorithm}[H]
591 \caption{Soma dos elementos de um vetor}
592 \label{alg:soma}
593 \Dados{vetor  $v$  com  $n$  elementos}
594 \Resultado{soma  $s$  dos elementos de  $v$ }
595  $s \leftarrow 0$ ;
596 \Para{ $i \leftarrow 1$  \Ate{  $n$  }}{
597      $s \leftarrow s + v[i]$ ;
598 }
599 \Retorna  $s$ ;
600 \end{algorithm}

```

```

601
602 \chapter{Método Proposto}
603 \chapter{Resultados e Discuss~oes}
604
605 \section{Algumas Demonstra{\c c}\~oes}
606
607 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alph
608 \syml[beta]{Beta}{A palavra Beta more e corrigida}
609 \syml[zbeta]{\beta}{A letra $\beta$ corrigida}
610 \syml{beta}{A palavra beta}
611 \syml{alpha}{A palavra alpha}
612 \syml[alpha]{Alpha}{A palavra Alpha}
613 \syml[zalpha]{\alpha}{A letra $\alpha$ corrigida}
614 \syml[marco]{Marco}{A palavra Marco corrigida}
615
616 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exempl
617 \abbrev{GoT}{Game of Thrones}
618 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
619 \abbrev[iot]{IoT}{IoT ordenado como iot}
620 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
621 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
622 \abbrev{IoT}{IoT com ordenação default}
623 \abbrev[ITU]{ITU}{ITU mesmo}
624
625 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
626 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
627
628 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas auto
629
630
631
632
633 \chapter{Conclusões}
634
635 \backmatter
636
637 \printbibliography
638
639
640
641 \appendix
642
643 \chapter{Um apêndice}
644
645 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
646
647 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo
648
649
650 \chapter{Outro apêndice}
651
652 \annex
653
654

```

```

655 \chapter{Um Anexo}
656 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
657
658
659
660 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
661
662 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comand
663
664
665 \chapter{Outro Anexo}
666
667
668 \end{document}
669
670 </example>

```

8 Implementation

8.1 The ‘coppe.cls’ file

```

671 <*class>
672 \def\filename{coppe.dtx}
673 \def\fileversion{v3.8}
674 \def\filedate{2026/05/25}
675 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
676 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
677 \LoadClass[12pt,a4paper,twoside]{book}
678 % The bibliography engine (biblatex+biber) is loaded further below, after babel,
679 % so the numbers class option can select the citation style at load time.
680 \RequirePackage{hyphenat}
681 \RequirePackage{lastpage}
682 \RequirePackage{ifthen}
683 \RequirePackage{graphicx}
684 \RequirePackage{setspace}
685 \RequirePackage{tabularx}
686 \RequirePackage{etoolbox}
687 \RequirePackage{eqparbox}
688 \RequirePackage{ltxcmds}
689 \RequirePackage[T1]{fontenc}
690 % manual2024 margins: 3cm top, 2cm bottom, mirrored 3cm inner / 2cm outer for
691 % two-sided printing -- realised as inner=outer=2cm plus a 1cm bindingoffset.
692 \RequirePackage[a4paper,twoside,bindingoffset=1cm,%
693 top=3cm,bottom=2cm,inner=2cm,outer=2cm]{geometry}
694 \RequirePackage{titlesec}
695 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
696 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capítulo N").
697 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
698 % keep "N Title". \appendix and \annex raise the flag (see below).
699 \newif\if@coppeappendix \@coppeappendixfalse
700 \titleformat{\chapter}[hang]
701   {\normalfont\Large\bfseries}
702   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
703   {1ex}\MakeUppercase}

```

```

704 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
705 \let\coppe@orig@appendix\appendix
706 \renewcommand\appendix{\coppe@orig@appendix\@coppeappendixtrue}
707 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
708 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
709 \titleformat{name=\chapter,numberless}[hang]
710 {\normalfont\Large\bfseries\filcenter}{0pt}{0pt}{\MakeUppercase}
711 \titlespacing*{name=\chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
712 % section: ALL CAPS, normal weight.
713 \titleformat{section}
714 {\normalfont\large\bfseries}{\thesection}{1ex}{\MakeUppercase}
715 % subsection: bold (author writes Title Case).
716 \titleformat{subsection}
717 {\normalfont\bfseries}{\thesubsection}{1ex}{}
718 % subsubsection: bold italic.
719 \titleformat{subsubsection}
720 {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{}
721 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
722 % "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
723 \RequirePackage{caption}
724 \captionsetup{font=footnotesize,labelsep=endash}
725 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
726 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
727 % at the bottom is mandatory). The float package's plaintop style forces the
728 % caption above the body for figure, table and the new quadro float. -----
729 \RequirePackage{float}
730 \floatstyle{plaintop}
731 \restylefloat{figure}
732 \restylefloat{table}
733 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
734 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
735 % unlike a tabela (rules on top and bottom only).
736 \providecommand{\quadroname}{\iflanguage{brazilian}{Quadro}{Frame}}
737 \providecommand{\listquadroname}{%
738 \iflanguage{brazilian}{Lista de Quadros}{List of Frames}}
739 \newfloat{quadro}{htbp}{loq}[chapter]
740 \floatname{quadro}{\quadroname}
741 \providecommand{\quadroautorefname}{\quadroname}
742 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
743 % (float's own \l@quadro lays the number/title out incorrectly).
744 \let\l@quadro\l@figure
745 \newcommand\listofquadros{%
746 \chapter*{\listquadroname}%
747 \addcontentsline{toc}{chapter}{\listquadroname}%
748 \@mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
749 \@starttoc{loq}}
750 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
751 % float as a smaller, single-spaced, centred line; it does NOT advance any
752 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
753 % the label is "Fonte" (pt) / "Source" (en).
754 \newcommand{\cpsourcenname}{\iflanguage{brazilian}{Fonte}{Source}}
755 \newcommand{\cpsource}[1]{%
756 \par\addvspace{\smallskipamount}%
757 {\footnotesize\singlespacing\centering\cpsourcenname: #1\par}}

```

```

758 \let\cpfonte\cpsource
759 \@ifundefined{source}{\let\source\cpsource}{}%
760 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
761 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
762 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
763 \newcommand{\newcoppefloat}[3]{%
764   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
765   \floatname{#1}{#2}%
766   \expandafter\newcommand\csname listof#1s\endcsname{%
767     \chapter*{#3}\addcontentsline{toc}{chapter}{#3}%
768     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
769     \@starttoc{lo#1}}}%
770 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
771 % (filete) from the left margin; footnotesize is the book default. -----
772 \let\coppe@orig@footnotetext\@footnotetext
773 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
774 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
775 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
776 % (manual2024 2.6 / proposta P24). -----
777 \RequirePackage{enumitem}
778 \newlist{alineas}{enumerate}{2}
779 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
780 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
781 % --- Page numbering: arabic in the top OUTER corner on textual pages
782 % (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
783 % \frontmatter (unnumbered by default; roman with the numeraisromanos
784 % class option). -----
785 \RequirePackage{fancyhdr}
786 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
787 \renewcommand{\headrulewidth}{0pt}
788 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
789 % number may leak onto them (e.g. the verso right after the cover).
790 \renewcommand{\cleardoublepage}{%
791   \clearpage
792   \if@twoside
793     \ifodd\c@page\else
794       \hbox{}\thispagestyle{empty}\newpage
795       \if@twocolumn\hbox{}\newpage\fi
796     \fi
797   \fi}
798 \def\CoppeTeX{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
799 p\kern-.025em e}}\kern-.08em
800 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}

801 \newboolean{maledoc}
802 \setboolean{maledoc}{false}
803 %
804 % Class options.
805 % If you are writing a text in English, you must turn ‘English’ on.
806 % Otherwise, Portuguese is considered the main language.
807 \newif\if@english\@englishfalse
808 \DeclareOption{english}{\@englishtrue}
809 \DeclareOption{msc}{%
810   \newcommand{\@degree}{M.Sc.}
811   \newcommand{\@degreename}{Mestrado}

```

```

812 \newcommand{\local@degname}{Mestre}
813 \newcommand{\foreign@degname}{Master}
814 \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
815 \newcommand\foreign@doctype{Dissertation}
816 }
817 \DeclareOption{dsccexam}{%
818 \newcommand{\@degree}{D.Sc.}
819 \newcommand{\@degreename}{Doutorado}
820 \newcommand{\local@degname}{Doutor}
821 \newcommand{\foreign@degname}{Doctor}
822 \setboolean{maledoc}{true}
823 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
824 \newcommand\foreign@doctype{Qualifying Exam}
825 }
826 \DeclareOption{msccexam}{%
827 \newcommand{\@degree}{M.Sc.}
828 \newcommand{\@degreename}{Mestrado}
829 \newcommand{\local@degname}{Mestre}
830 \newcommand{\foreign@degname}{Master}
831 \setboolean{maledoc}{true}
832 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
833 \newcommand\foreign@doctype{Qualifying Exam}
834 }
835 \DeclareOption{dsc}{%
836 \newcommand{\@degree}{D.Sc.}
837 \newcommand{\@degreename}{Doutorado}
838 \newcommand{\local@degname}{Doutor}
839 \newcommand{\foreign@degname}{Doctor}
840 \newcommand\local@doctype{Tese}
841 \newcommand\foreign@doctype{Thesis}
842 }
843 \newif\if@coppenumeric\@coppenumericfalse
844 \DeclareOption{numbers}{\@coppenumerictrue}
845 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
846 % pre-textual pages are counted but not numbered (manual2024 2.7).
847 \newif\if@copperoman\@copperomanfalse
848 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Here is the default one-and-a-half line spacing. Users can change to double spacing by passing the `doublespacing` option.

```

849 \onehalfspacing
850 \DeclareOption{doublespacing}{%
851 \doublespacing
852 }
853 \ProcessOptions\relax
854 \if@english
855 \RequirePackage[brazilian,english]{babel}
856 \else
857 \RequirePackage[english,brazilian]{babel}
858 \fi
859 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
860 % Portuguese alias for short, inline quotations (proposta P3).
861 \RequirePackage{csquotes}
862 \newcommand{\citacao}[1]{\enquote{#1}}

```

```

863 % --- Code listings & algorithms (TOD03 "tratar listagens"): the listings
864 %     package gives the "Programa" listing (caption on top, ABNT R46) with
865 %     \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
866 %     the language styles below are based on the user's listagens.tex. -----
867 \RequirePackage{xcolor}
868 \RequirePackage{listings}
869 \RequirePackage{fancyvrb}
870 \definecolor{deepblue}{rgb}{0,0,0.5}
871 \definecolor{deepred}{rgb}{0.6,0,0}
872 \definecolor{deepgreen}{rgb}{0,0.5,0}
873 \DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
874 \DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
875 \newcommand{\gxnumbersep}{0.5em}
876 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
877   literate=%
878   {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{e}}1%
879   {\~}{\~{e}}1 {\~}{\~{e}}1 {\~}{\~{i}}1 {\~}{\~{i}}1 {\~}{\~{o}}1%
880   {\~}{\~{o}}1 {\~}{\~{o}}1 {\~}{\~{u}}1 {\~}{\~{u}}1 {\~}{\~{c}}1%
881   {\~}{\~{c}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{E}}1%
882   {\~}{\~{E}}1 {\~}{\~{I}}1 {\~}{\~{I}}1 {\~}{\~{O}}1 {\~}{\~{O}}1 {\~}{\~{O}}1%
883   {\~}{\~{U}}1%
884 }
885 % Babel-aware listing names, set after listings' own babel setup.
886 \AtBeginDocument{%
887   \renewcommand{\lstlistingname}{\iflanguage{brazilian}{Programa}{Program}}%
888   \renewcommand{\lstlistlistingname}{%
889     \iflanguage{brazilian}{Lista de Programas}{List of Programs}}%
890 % \listofprogramas in the coppe list style (heading + ToC + running head).
891 \newcommand{\listofprogramas}{%
892   \chapter*{\lstlistlistingname}%
893   \addcontentsline{toc}{chapter}{\lstlistlistingname}%
894   \@mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
895   \@starttoc{lol}}
896 % Per-language styles + environments (python, java, xml, html, prolog).
897 \newcommand{\pythonstyle}{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
898   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
899   morekeywords={self,as},keywordstyle=\ttb\color{deepblue},%
900   emph={MyClass,__init__},emphstyle=\ttb\color{deepred},%
901   stringstyle=\color{deepgreen},frame=tb,showstringspaces=false,%
902   extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
903   escapechar=',postbreak=\mbox{\textcolor{red}{\$}\hookrightarrow}\space}}
904 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}%
905   \csname\@lst @SetFirstNumber\endcsname]{\csname\@lst @SaveFirstNumber\endcsname}
906 \newcommand{\pythoninline}[1][\pythonstyle\lstinline!#1!}%
907 \newcommand{\pythonexternal}[2][\pythonstyle\lstinputlisting{#1}{#2}}%
908 \newcommand{\xmlstyle}{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
909   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,%
910   morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
911   keywordstyle=\ttb\color{deepblue},%
912   emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
913   emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
914   showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
915   postbreak=\mbox{\textcolor{red}{\$}\hookrightarrow}\space}}
916 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]{%

```

```

917 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
918   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,%
919   keywordstyle=\ttb\color{deepblue},emphstyle=\ttb\color{deepred},%
920   stringstyle=\color{deepgreen},frame=tb,showstringspaces=false,%
921   extendedchars=true,inputencoding=utf8,breaklines=true,%
922   postbreak=\mbox{\textcolor{red}{\hookrightarrow}\space}}}
923 \lstnewenvironment{html}[1][\htmlstyle\lstset{#1}]{%
924 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
925   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
926   keywordstyle=\ttb\color{deepblue},emph={MyClass, __init__},%
927   emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
928   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
929   breaklines=true,escapechar=',%
930   postbreak=\mbox{\textcolor{red}{\hookrightarrow}\space}}}%
931 \lstnewenvironment{prolog}[1][\prologstyle\lstset{#1}%
932   \csname\@lst @SetFirstNumber\endcsname\{\csname\@lst @SaveFirstNumber\endcsname\}
933 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
934   numberstyle=\small\ttfamily,numbersep=\gxnumbersep,commentstyle=\color{purple},%
935   morekeywords={self,as},keywordstyle=\ttb\color{deepblue},%
936   emphstyle=\ttb\color{deepred},stringstyle=\color{deepgreen},frame=tb,%
937   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
938   breaklines=true,escapechar=',%
939   postbreak=\mbox{\textcolor{red}{\hookrightarrow}\space}}}%
940 \lstnewenvironment{java}[1][\javastyle\lstset{#1}%
941   \csname\@lst @SetFirstNumber\endcsname\{\csname\@lst @SaveFirstNumber\endcsname\}
942 \newcommand\javainline[1][\javastyle\lstinline!#1!}%
943 \newcommand\javaexternal[2][\javastyle\lstinputlisting[1]{#2}]}
944 % Verbatim output environments (program output), fancyvrb.
945 \DefineVerbatimEnvironment{gxoutput}{Verbatim}%
946   {frame=lines,numbers=left,numbersep=\gxnumbersep}
947 \DefineVerbatimEnvironment{gxoutputs}{Verbatim}%
948   {frame=lines,numbers=left,numbersep=\gxnumbersep,fontsize=\small}
949 \renewcommand\theFancyVerbLine{\small\ttfamily\arabic{FancyVerbLine}}
950 % Algorithms: algorithm2e, language following the english class option.
951 \if@english
952   \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
953 \else
954   \RequirePackage[portuguese,linesnumbered,lined,algochapter,ruled]{algorithm2e}
955 \fi
956 \SetKwFor{Para}{para}{faça}{fim do para}
957 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
958 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
959 % former natbib+bibtex setup. Loaded here, after babel, so its language strings
960 % map correctly. The numbers option selects the numeric style; otherwise the
961 % author-date default is used. natbib=true keeps \citel/\citep available.
962 \if@coppenumeric
963   \RequirePackage[backend=biber,database=coppe,%
964     bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
965 \else
966   \RequirePackage[backend=biber,database=coppe,%
967     bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
968 \fi
969 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
970 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not

```



```

971 % "Bibliografia" (babel's default \bibname). The title is set directly with
972 % \iflanguage so it is correct in BOTH the chapter heading and the .toc entry:
973 % \iflanguage resolves to plain text and survives the write to the .toc, whereas
974 % biblatex's \bibstring does not, and babel's \bibname caption override proved
975 % unreliable here. Evaluated in the body, the current language is settled.
976 \defbibheading{bibliography}{%
977   \chapter*{\iflanguage{brazilian}{Refer\~encias}{References}}%
978   \addcontentsline{toc}{chapter}{\iflanguage{brazilian}{Refer\~encias}{References}}}
979 % The catalog page range (bib:begin/bib:end) was written by the old
980 % thebibliography redefinition, which biblatex's \printbibliography does not
981 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
982 % begin label uses that hook and the end label is emitted by wrapping
983 % \printbibliography to write it just after the reference list.
984 \AtBeginBibliography{\coppe@bibBegin}
985 \let\coppe@orig@printbibliography\printbibliography
986 \renewcommand\printbibliography[1][]{%
987   \coppe@orig@printbibliography[#1]%
988   \coppe@bibEnd}

```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

989 \newcommand\department[1]{%
990   \ifthenelse{\equal{#1}{PEB}}{
991     {\global\def\local@deptname{Engenharia Biom(\` e)dica}
992     \global\def\foreign@deptname{Biomedical Engineering}}{
993     \ifthenelse{\equal{#1}{PEC}}{
994       {\global\def\local@deptname{Engenharia Civil}
995       \global\def\foreign@deptname{Civil Engineering}}{
996       \ifthenelse{\equal{#1}{PEE}}{
997         {\global\def\local@deptname{Engenharia El(\` e)trica}
998         \global\def\foreign@deptname{Electrical Engineering}}{
999         \ifthenelse{\equal{#1}{PEM}}{
1000           {\global\def\local@deptname{Engenharia Mec(\` a)nica}
1001           \global\def\foreign@deptname{Mechanical Engineering}}{
1002         \ifthenelse{\equal{#1}{PEMM}}{
1003           {\global\def\local@deptname{Engenharia Metal(\` u)rgica e de Materiais}
1004 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{
1005         \ifthenelse{\equal{#1}{PEN}}{
1006           {\global\def\local@deptname{Engenharia Nuclear}
1007           \global\def\foreign@deptname{Nuclear Engineering}}{
1008         \ifthenelse{\equal{#1}{PEN0}}{
1009           {\global\def\local@deptname{Engenharia Oce(\` a)nica}
1010           \global\def\foreign@deptname{Ocean Engineering}}{
1011         \ifthenelse{\equal{#1}{PPE}}{
1012           {\global\def\local@deptname{Planejamento Energ(\` e)tico}
1013           \global\def\foreign@deptname{Energy Planning}}{
1014         \ifthenelse{\equal{#1}{PEP}}{
1015           {\global\def\local@deptname{Engenharia de Produ(c)c(\` a)o}
1016           \global\def\foreign@deptname{Production Engineering}}{
1017         \ifthenelse{\equal{#1}{PEQ}}{
1018           {\global\def\local@deptname{Engenharia Qu(\` i)mica}
1019           \global\def\foreign@deptname{Chemical Engineering}}{
1020         \ifthenelse{\equal{#1}{PESC}}{

```

```

1021     {\global\def\local@deptname{Engenharia de Sistemas e Computa{\c c}{\~ a}o}
1022     \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
1023 \ifthenelse{\equal{#1}{PET}}
1024     {\global\def\local@deptname{Engenharia de Transportes}
1025     \global\def\foreign@deptname{Transportation Engineering}}{}
1026 \ifthenelse{\equal{#1}{PENT}}
1027     {\global\def\local@deptname{Engenharia de Nanotecnologia}
1028     \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1029 }

\title Used to enter the title in Brazilian Portuguese.
1030 \renewcommand\title[1]{%
1031     \global\def\local@title{#1}%
1032 }

\foreigntitle Used to enter the foreign title.
1033 \newcommand\foreigntitle[1]{%
1034     \global\def\foreign@title{#1}%
1035 }

\advisor Defines globally the title, name and academic degree of the advisors.
1036 \newcount\@advisor\@advisor0
1037 \newcommand\advisor[4]{%
1038     \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
1039     \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
1040     \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
1041     \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
1042     \global\advance\@advisor by 1
1043     \ifnum\@advisor>1
1044         \renewcommand\local@advisorstring{Orientadores}
1045         \renewcommand\foreign@advisorstring{Advisors}
1046     \fi
1047 }

\examiner
1048 \newcount\@examiner\@examiner0
1049 \newcommand\examiner[3]{%
1050     \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
1051     \global\advance\@examiner by 1
1052 }

\author It was redefined to allow the identification of the author's first names and surname.
1053 \renewcommand\author[2]{%
1054     \global\def\@authname{#1}
1055     \global\def\@authsurn{#2}
1056 }

\date This code makes easy to switch from dates in different languages.
1057 \renewcommand\date[2]{%
1058     \month=#1
1059     \year=#2
1060 }

```

`\local@monthname`

```
1061 \newcommand\local@monthname{\ifcase\month\or
1062   Janeiro\or Fevereiro\or Mar{\c c}o\or Abril\or Maio\or Junho\or
1063   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}
```

`\foreign@monthname`

```
1064 \newcommand\foreign@monthname{\ifcase\month\or
1065   January\or February\or March\or April\or May\or June\or
1066   July\or August\or September\or October\or November\or December\fi}
```

`\keyword`

```
1067 \newcounter{keywords}
1068 \newcommand\keyword[1]{%
1069   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
1070   \global\addtocounter{keywords}{1}
1071 }
```

`\freeconfig` This command allows easy changing of core class parameters.

```
1072 \newcommand\freeconfig[6]{%
1073   \providecommand\@degree{}
1074   \renewcommand\@degree{#1}
1075   \providecommand\@degreename{}
1076   \renewcommand\@degreename{#2}
1077   \providecommand\local@degname{}
1078   \renewcommand\local@degname{#3}
1079   \providecommand\foreign@degname{}
1080   \renewcommand\foreign@degname{#4}
1081   \providecommand\local@doctype{}
1082   \renewcommand\local@doctype{#5}
1083   \providecommand\foreign@doctype{}
1084   \renewcommand\foreign@doctype{#6}%
1085 }
1086 % \end{macrocode}
1087 % \end{macro}
1088 %
1089 % \begin{macro}{\frontmatter}
1090 % The number of pages for both frontmatter and mainmatter printed
1091 % in the cataloging details page is computed by means of simple
1092 % \LaTeX\ labels.
1093 % \begin{macrocode}
1094 \renewcommand\frontmatter{%
1095   \cleardoublepage
1096   \@mainmatterfalse
1097   \pagenumbering{roman}
1098   \thispagestyle{empty}
1099   % folha de rosto = counted page 1 (ABNT: the capa is not counted). An odd
1100   % start keeps the page-counter parity aligned with the physical (recto) side,
1101   % so \cleardoublepage lands every pre-textual element on an odd (front) page.
1102   \setcounter{page}{1}
1103   \makefrontpage
1104   \clearpage
1105   \if@copperoman
1106     \fancypagestyle{plain}{\fancyhf{} \fancyhead[R0,LE]{\thepage}}%
1107     \renewcommand{\headrulewidth}{0pt}}%
```

```

1108     \pagestyle{coppe}%
1109   \else
1110     \fancypagestyle{plain}{\fancyhf{}}\renewcommand{\headrulewidth}{0pt}}%
1111     \pagestyle{empty}%
1112   \fi
1113   \ifthenelse{\boolean{maledoc}}{\}{\makecatalog}%
1114 }

\mainmatter
1115 \renewcommand\mainmatter{%
1116   \coppe@mainBegin
1117   \cleardoublepage
1118   \@mainmattertrue
1119   \fancypagestyle{plain}{\fancyhf{}}\fancyhead[R0,LE]{\thepage}%
1120   \renewcommand{\headrulewidth}{0pt}}%
1121   \pagestyle{coppe}%
1122   \pagenumbering{arabic}}

\backmatter
1123 \renewcommand\backmatter{%
1124   \if@openright
1125     \cleardoublepage
1126   \else
1127     \clearpage
1128   \fi}
1129 %

\maketitle
1130 \renewcommand\maketitle{%
1131   \pagenumbering{alph}
1132   \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
1133   \begin{titlepage}
1134     \begin{flushleft}
1135       \vspace*{1.5mm}
1136       \setlength\baselineskip{0pt}
1137       \setlength\parskip{1mm}
1138       \makebox[\linewidth]{%
1139         \includegraphics[height=2cm]{ufrj-logo}\hfill
1140         \includegraphics[height=2cm]{coppe-logo}}
1141       \end{flushleft}
1142       \vspace{1.05cm}
1143       \begin{center}
1144         \nohyphens{%
1145           \if@english
1146             \MakeUppercase\foreign@title
1147           \else
1148             \MakeUppercase\local@title
1149           \fi}\par
1150         \vspace*{3cm}
1151         \nohyphens{\@authname\ \@authsurn}\par
1152       \end{center}
1153       \vspace*{2.1cm}
1154       \begin{flushright}
1155         \begin{minipage}{8.45cm}

```

```

1156 \frontcover@maintext
1157 \end{minipage}\par
1158 \vspace*{7.5mm}
1159 \nohyphens{%
1160 \begin{tabularx}{8.45cm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1161   \local@advisorstring: &
1162   \count1=0
1163   \toks@={}
1164   \@whilenum \count1<\@advisor \do{%
1165     \ifcase\count1 % same as \ifnum0=\count1
1166       \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1167         \expandafter\endcsname\expandafter\space%
1168         \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1169     \else
1170       \toks@=\expandafter\expandafter\expandafter{%
1171         \expandafter\the\expandafter\toks@%
1172         \expandafter&\expandafter\space%
1173         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1174         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1175       }%
1176     \fi
1177     \advance\count1 by 1}
1178   \the\toks@
1179 \end{tabularx}}\par
1180 \end{flushright}
1181 \vspace*{\fill}
1182 \begin{center}
1183 \local@cityname\par
1184 \local@monthname\ de \number\year
1185 \end{center}
1186 \end{titlepage}
1187 \global\let\maketitle\relax%
1188 \global\let\and\relax}

1189 \newcommand\makefrontpage{%
1190   \begin{center}
1191     \sloppy\nohyphens{
1192       \if@english
1193         \MakeUppercase\foreign@title
1194       \else
1195         \MakeUppercase\local@title
1196       \fi}\par
1197     \vspace*{7mm}
1198     {\@authname\ \@authsurn}\par
1199   \end{center}\par
1200   \vspace*{4mm}
1201   \frontpage@maintext
1202   \vspace*{16mm}
1203   \nohyphens{%
1204     \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1205       \local@advisorstring: &
1206       \count1=0
1207       \toks@={}
1208       \@whilenum \count1<\@advisor \do{%

```

```

1209 \ifcase\count1 % same as \ifnum0=\count1
1210 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1211 \expandafter\endcsname\expandafter\space%
1212 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1213 \else
1214 \toks@=\expandafter\expandafter\expandafter{%
1215 \expandafter\the\expandafter\toks@%
1216 \expandafter&\expandafter\space%
1217 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1218 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1219 }%
1220 \fi
1221 \advance\count1 by 1}
1222 \the\toks@
1223 \end{tabularx}\par
1224 \vspace*{20mm}
1225 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1226 Aprovada por: &
1227 \count1=0
1228 \toks@={}
1229 \@whilenum \count1<\@examiner \do{%
1230 \ifcase\count1 % same as \ifnum0=\count1
1231 \toks@=\expandafter{\csname CoppeExaminer:\the\count1%
1232 \expandafter\endcsname\expandafter\\}
1233 \else
1234 \toks@=\expandafter\expandafter\expandafter{%
1235 \expandafter\the\expandafter\toks@%
1236 \expandafter&\expandafter\space%
1237 \csname CoppeExaminer:\the\count1\expandafter\endcsname%
1238 \expandafter\space\\
1239 }%
1240 \fi
1241 \advance\count1 by 1}
1242 \the\toks@
1243 \end{tabularx}}\par
1244 \vspace*{\fill}
1245 \frontpage@bottomtext}

1246 \newcommand\coppe@hypersetup{%
1247 \begingroup
1248 % changes to \toks@ and \count@ are kept local;
1249 % it's not necessary for them, but it is usually the case
1250 % for \count1, because the first ten counters are written
1251 % to the DVI file, thus you got lucky because of PDF output
1252 \toks@={} % in this special case not necessary
1253 \count@=0 %
1254 \@whilenum\count@<\value{keywords}\do{%
1255 % * a keyword separator is not necessary,
1256 % if there is just one keyword
1257 % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
1258 % at least once, to get rid of the loop depended \count@
1259 \ifcase\count@ % same as \ifnum0=\count@
1260 \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
1261 \else

```

```

1262     \toks@=\expandafter\expandafter\expandafter{%
1263         \expandafter\the\expandafter\toks@
1264         \expandafter;\expandafter\space
1265         \cename CoppeKeyword:\the\count@\endcsname
1266     }%
1267     \fi
1268     \advance\count@ by 1 %
1269 }%
1270 \edef\x{\endgroup
1271     \noexpand\hypersetup{%
1272         pdfkeywords={\the\toks@}%
1273     }%
1274 }%
1275 \x
1276 \hypersetup{%
1277     pdfauthor={\@authname\ \@authsurn},
1278     pdftitle={\local@title},
1279     pdfsubject={\local@doctype\ de \@degreename\ em \local@deptname\ da COPPE/UFRJ},
1280     pdfcreator={LaTeX with CoppeTeX toolkit},
1281     breaklinks={true},
1282     raiselinks={true},
1283     pageanchor={true},
1284 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il.” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain T_EX command `\isundefined [?]`.

```

1285 \newcommand\makecatalog{%
1286     \vspace*{\fill}
1287     \begin{center}
1288         \setlength{\fboxsep}{5mm}
1289         \framebox[120mm][c]{\makebox[5mm][c]{}%
1290             \begin{minipage}[c]{105mm}
1291                 \setlength{\parindent}{5mm}
1292                 \noindent\sloppy\nohyphens\@authsurn,
1293                 \nohyphens\@authname\par
1294                 \nohyphens{%
1295                     \if@english
1296                         \foreign@title%
1297                     \else
1298                         \local@title%
1299                     \fi/\@authname\ \@authsurn. -- \local@cityname:
1300                     UFRJ/COPPE, \number\year.}\par
1301                 \pageref{front:pageno},
1302                 \pageref{LastPage}
1303                 p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm}.\par
1304                 % There is an issue here. When the last entry must be split between lines,
1305                 % the spacing between it and the next paragraph becomes smaller.
1306                 % Should we manually introduce a fixed space? But how could we know that
1307                 % a name was split? Is this happening yet?
1308                 \nohyphens{%
1309                     \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1310                         \local@advisorstring: &

```

```

1311 \count1=0
1312 \toks@={}
1313 \@whilenum \count1<\@advisor \do{%
1314 \ifcase\count1 % same as \ifnum0=\count1
1315 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1316 \expandafter\endcsname\expandafter\space%
1317 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1318 \else
1319 \toks@=\expandafter\expandafter\expandafter{%
1320 \expandafter\the\expandafter\toks@
1321 \expandafter&\expandafter\space
1322 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1323 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1324 }%
1325 \fi
1326 \advance\count1 by 1}
1327 \the\toks@
1328 \end{tabularx}}\par
1329 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
1330 UFRJ/COPPE/Programa de \local@deptname, \number\year.}\par
1331 Refer{\^ e}ncias Bibliogr{\' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
1332 \count1=0
1333 \count2=1
1334 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
1335 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
1336 \advance\count1 by 1
1337 \advance\count2 by 1}
1338 I. \csname CoppeAdvisorSurname:0\endcsname,%
1339 \ \csname CoppeAdvisorName:0\endcsname%
1340 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
1341 II. \local@universityname, COPPE, Programa de \local@deptname.
1342 III. T{\' i}tulo.}
1343 \end{minipage}}
1344 \end{center}
1345 \vspace*{\fill}}

```

\dedication

```

1346 \newcommand\dedication[1]{
1347 \gdef\@dedic{#1}
1348 \cleardoublepage
1349 \vspace*{\fill}
1350 \begin{flushright}
1351 \begin{minipage}{60mm}
1352 \raggedleft \it \normalsize \@dedic
1353 \end{minipage}
1354 \end{flushright}}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

1355 \newenvironment{abstract}{%
1356 \cleardoublepage
1357 \thispagestyle{plain}
1358 \abstract@toptext\par
1359 \vspace*{8.6mm}
1360 \begin{center}

```



```

1361 \sloppy\nohyphens{\MakeUppercase\local@title}\par
1362 \vspace*{13.2mm}
1363 \@authname\ \@authsur \par
1364 \vspace*{7mm}
1365 \local@monthname/\number\year
1366 \end{center}\par
1367 \vspace*{1.5cm}
1368 \noindent%
1369 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1370 \local@advisorstring: &
1371 \count1=0
1372 \toks@={}
1373 \@whilenum \count1<\@advisor \do{%
1374 \ifcase\count1 % same as \ifnum0=\count1
1375 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1376 \expandafter\endcsname\expandafter\space%
1377 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1378 \else
1379 \toks@=\expandafter\expandafter\expandafter{%
1380 \expandafter\the\expandafter\toks@
1381 \expandafter&\expandafter\space
1382 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1383 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1384 }%
1385 \fi
1386 \advance\count1 by 1}
1387 \the\toks@
1388 \end{tabularx}\par
1389 \vspace*{2mm}
1390 \noindent\local@deptstring: \local@deptname\par
1391 \vspace*{7mm}}{\vspace*{\fill}}

```

foreignabstract (env.)

```

1392 \newenvironment{foreignabstract}{%
1393 \cleardoublepage
1394 \thispagestyle{plain}
1395 \begin{otherlanguage}{english}
1396 \foreignabstract@toptext\par
1397 \vspace*{8.6mm}
1398 \begin{center}
1399 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
1400 \vspace*{13.2mm}
1401 \@authname\ \@authsur \par
1402 \vspace*{7mm}
1403 \foreign@monthname/\number\year
1404 \end{center}\par
1405 \vspace*{1.5cm}
1406 \noindent%
1407 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1408 \foreign@advisorstring: &
1409 \count1=0
1410 \toks@={}
1411 \@whilenum \count1<\@advisor \do{%
1412 \ifcase\count1 % same as \ifnum0=\count1

```

```

1413 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1414 \expandafter\endcsname\expandafter\space%
1415 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1416 \else
1417 \toks@=\expandafter\expandafter\expandafter{%
1418 \expandafter\the\expandafter\toks@
1419 \expandafter&\expandafter\space
1420 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1421 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1422 }%
1423 \fi
1424 \advance\count1 by 1}
1425 \the\toks@
1426 \end{tabularx}\par
1427 \vspace*{2mm}
1428 \noindent\foreign@deptstring: \foreign@deptname\par
1429 \vspace*{7mm}}{%
1430 \end{otherlanguage}
1431 \vspace*{\fill}
1432 \global\let\@author\@empty
1433 \global\let\@date\@empty
1434 \global\let\foreign@title\@empty
1435 \global\let\foreign@title\relax
1436 \global\let\local@title\@empty
1437 \global\let\local@title\relax
1438 \global\let\author\relax
1439 \global\let\author\relax
1440 \global\let\date\relax}

```

\listoffigures

```

1441 \renewcommand\listoffigures{%
1442 \coppe@hasLof
1443 \if@twocolumn
1444 \@restonecoltrue\onecolumn
1445 \else
1446 \@restonecolfalse
1447 \fi
1448 \chapter*{\listfigurename}%
1449 \addcontentsline{toc}{chapter}{\listfigurename}%
1450 \@mkboth{\MakeUppercase\listfigurename}%
1451 {\MakeUppercase\listfigurename}%
1452 \@starttoc{lof}%
1453 \if@restonecol\twocolumn\fi
1454 }

```

\listoftables

```

1455 \renewcommand\listoftables{%
1456 \if@twocolumn
1457 \@restonecoltrue\onecolumn
1458 \else
1459 \@restonecolfalse
1460 \fi
1461 \chapter*{\listtablename}%
1462 \addcontentsline{toc}{chapter}{\listtablename}%

```

```

1463      \@mkboth{%
1464          \MakeUppercase\listtablename}%
1465          {\MakeUppercase\listtablename}%
1466      \@starttoc{lot}%
1467      \if@restonecol\twocolumn\fi
1468      }

\printlosymbols

1469 \newcommand\printlosymbols{%
1470 \renewcommand\glossaryname{\listsymbolname}%
1471 \@input@{\jobname.los}}

\makelosymbols

1472 \def\makelosymbols{%
1473   \newwrite\@losfile
1474   \immediate\openout\@losfile=\jobname.syx
1475   \newcommand\syml[3][\@bsphack\beginingroup
1476   \ifstrepty{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
1477       \@sanitize%
1478       \@wrls{\@tempsymb1}{##2}{##3}}\typeout%
1479   {Writing index of symbols file \jobname.syx}%
1480   \let\makelosymbols\@empty%
1481 }%
1482 \@onlypreamble\makelosymbols

1483 \AtBeginDocument{%
1484 \@ifpackageloaded{hyperref}{%
1485   \newcommand\@wrls[3]{%
1486     \protected@write\@losfile{%
1487       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
1488     \endgroup%
1489     \@esphack}}{%
1490   \newcommand\@wrls[3]{%
1491     \protected@write\@losfile{%
1492       {\string\indexentry{#1[#2] #3}{\thepage}}%
1493     \endgroup%
1494     \@esphack}}}%

\printloabbreviations

1495 \newcommand\printloabbreviations{%
1496 \renewcommand\glossaryname{\listabbreviationname}%
1497 \@input@{\jobname.lab}}

\makeloabbreviations

1498 \def\makeloabbreviations{%
1499   \newwrite\@labfile
1500   \immediate\openout\@labfile=\jobname.abx
1501   \newcommand\abbrev[3][\@bsphack\beginingroup
1502   \ifstrepty{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
1503       \@sanitize
1504       \@wrlab{\@tempsymb1}{##2}{##3}}\typeout
1505   {Writing index of abbreviations file \jobname.abx}%
1506   \let\makeloabbreviations\@empty
1507 }
1508 \@onlypreamble\makeloabbreviations

```

```

1509 \AtBeginDocument{%
1510 \ifpackageloaded{hyperref}{%
1511   \newcommand\wrlab[3]{%
1512     \protected@write\@labfile{%
1513       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
1514     \endgroup%
1515     \@esphack}}{%
1516   \newcommand\wrlab[3]{%
1517     \protected@write\@labfile{%
1518       {\string\indexentry{#1[#2] #3}{\thepage}}%
1519     \endgroup%
1520     \@esphack}}}%

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
\sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in
the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations
must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full
form. The first-use flag is set globally so it survives groups (floats, footnotes).

1521 \newcommand\newsigla[3]{%
1522   \@namedef{coppe@sigla@s@#1}{#2}%
1523   \@namedef{coppe@sigla@l@#1}{#3}%
1524   \newcommand\coppe@sigla@reg[1]{%
1525     \ifundefined{abbrev}{}%
1526     {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}%
1527   \newcommand\coppe@sigla@full[1]{%
1528     \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}
1529   \newcommand\coppe@sigla@undef[1]{%
1530     \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
1531     \textbf{???#1???}}
1532   \newcommand\coppe@sigla@first[1]{%
1533     \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
1534   \newcommand\coppe@sigla@norm[1]{%
1535     \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
1536     {\@ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
1537      {\@nameuse{coppe@sigla@s@#1}}}%
1538   \newcommand\coppe@sigla@star[1]{%
1539     \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
1540     {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}%
1541   \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

1542 %%% \AtBeginDocument{%
1543 %%% \ifpackageloaded{hyperref}{%
1544 %%%   \def\wrlab#1#2{%
1545 %%%     \protected@write\@labfile{%
1546 %%%       {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
1547 %%%     \endgroup
1548 %%%     \@esphack}}{%
1549 %%%   \def\wrlab#1#2{%
1550 %%%     \protected@write\@labfile{%
1551 %%%       {\string\indexentry{[#1] #2}{\arabic{page}}}%
1552 %%%     \endgroup
1553 %%%     \@esphack}}}%

1554 % Some macros used to generate cataloging information.
1555 \AtBeginDocument{%

```

```

1556 \ltx@ifpackageloaded{hyperref}{
1557   \def\coppe@bibEnd{%
1558     \immediate\write\@auxout{%
1559       \string\newlabel{bib:end}{\arabic{page}}{page.\arabic{page}}}%
1560   \def\coppe@bibBegin{%
1561     \immediate\write\@auxout{%
1562       \string\newlabel{bib:begin}{\arabic{page}}{page.\arabic{page}}}%
1563   \def\coppe@mainBegin{%
1564     \immediate\write\@auxout{%
1565       \string\newlabel{front:pageno}{\Roman{page}}{page.\roman{page}}}%
1566   \def\coppe@hasLof{%
1567     \immediate\write\@auxout{%
1568       \string\newlabel{cat:lofflag}{:~il.;}{page.\roman{page}}}%
1569 }{%
1570   \def\coppe@bibEnd{%
1571     \immediate\write\@auxout{%
1572       \string\newlabel{bib:end}{\arabic{page}}}%
1573   \def\coppe@bibBegin{%
1574     \immediate\write\@auxout{%
1575       \string\newlabel{bib:begin}{\arabic{page}}}%
1576   \def\coppe@mainBegin{%
1577     \immediate\write\@auxout{%
1578       \string\newlabel{front:pageno}{\Roman{page}}}%
1579   \def\coppe@hasLof{%
1580     \immediate\write\@auxout{%
1581       \string\newlabel{cat:lofflag}{:~il.;}}%
1582 }%
1583 }
1584 % The reference list is now produced by biblatex's \printbibliography (see the
1585 % bibliography setup after babel, above). The former thebibliography
1586 % redefinition and its \bibindent are no longer used and have been removed; the
1587 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
1588 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (env.)

```

1589 \newlength{\recuolongquote}%
1590 \setlength{\recuolongquote}{4cm}%
1591 \newenvironment*{longquote}[1][default]{%
1592   \list{}%
1593   \footnotesize%
1594   \addtolength{\leftskip}{\recuolongquote}%
1595   \item[]%
1596   \singlespacing%
1597   \ifthenelse{\not\equal{#1}{default}}{\itshape\selectlanguage{#1}}{}%
1598 }{\endlist}%

1599 \newenvironment{theglossary}{%
1600   \if@twocolumn%
1601     \@restonecoltrue\onecolumn%
1602   \else%
1603     \@restonecolfalse%
1604   \fi%
1605   \@mkboth{\MakeUppercase\glossaryname}%
1606   {\MakeUppercase\glossaryname}%

```

```

1607 \chapter*{\glossaryname}%
1608 \addcontentsline{toc}{chapter}{\glossaryname}
1609 \list{}
1610 {\setlength{\listparindent}{0in}%
1611 \setlength{\labelwidth}{1.0in}%
1612 \setlength{\leftmargin}{1.5in}%
1613 \setlength{\labelsep}{0.5in}%
1614 \setlength{\itemindent}{0in}}%
1615 \sloppy}%
1616 {\if@restonecol\twocolumn\fi%
1617 \endlist}
1618 %
1619 \renewenvironment{theindex}{%
1620 \if@twocolumn
1621 \@restonecolfalse
1622 \else
1623 \@restonecoltrue
1624 \fi
1625 \twocolumn[\@makeschapterhead{\indexname}]%
1626 \mkboth{\MakeUppercase\indexname}%
1627 {\MakeUppercase\indexname}%
1628 \thispagestyle{plain}\parindent\z@
1629 \addcontentsline{toc}{chapter}{\indexname}
1630 \parskip\z@ \@plus .3\p@\relax
1631 \columnseprule \z@
1632 \columnsep 35\p@
1633 \let\item\@idxitem}
1634 {\if@restonecol\onecolumn\else\clearpage\fi}
1635 \if@english
1636 \newcommand\listabbreviationname{List of Abbreviations and Acronyms}
1637 \newcommand\listsymbolname{List of Symbols}
1638 \newcommand\glossaryname{Glossary}
1639 \else
1640 \newcommand\listabbreviationname{Lista de Abreviaturas e Siglas}
1641 \newcommand\listsymbolname{Lista de Símbolos}
1642 \newcommand\glossaryname{Glossário}
1643 \fi
1644 %
1645 \newcommand\local@advisorstring{Orientador}
1646 \newcommand\foreign@advisorstring{Advisor}
1647 \ifthenelse{\boolean{maledoc}}{%
1648 \newcommand\local@approvedname{Examinado por}%
1649 }{%
1650 \newcommand\local@approvedname{Examinada por}%
1651 }
1652 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
1653 \newcommand\local@deptstring{Programa}
1654 \newcommand\foreign@deptstring{Department}
1655 \newcommand\local@cityname{Rio de Janeiro}
1656 \newcommand\local@statename{RJ}
1657 \newcommand\local@countryname{Brasil}
1658 %
1659 \newcommand\frontcover@maintext{
1660 \sloppy\nohyphens{\local@doctype\ de \@degree\

```

```

1661 \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
1662 ao Programa de P{\ ' o}s-gradua{\c c}{\~ a}o em \local@deptname,
1663 COPPE, da \local@universityname, como parte dos requisitos
1664 necess{\ ' a}rios {\ ' a} obten{\c c}{\~ a}o do t{\ ' i}tulo de
1665 \local@degname\ em \local@deptname.}
1666 }
1667 %
1668 \newcommand\frontpage@maintext{
1669 \noindent {\MakeUppercase\local@doctype}
1670 \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
1671 \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
1672 DE P{\ ' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
1673 UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
1674 NECESS{\ ' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
1675 {\MakeUppercase\local@degname} EM CI{\^ E}NCIAS EM
1676 {\MakeUppercase\local@deptname.\par}}%
1677 }
1678 %
1679 \newcommand\frontpage@bottomtext{%
1680 \begin{center}
1681 {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
1682 {\MakeUppercase\local@monthname\ DE \number\year}
1683 \end{center}%
1684 }
1685 %
1686 \newcommand\abstract@toptext{%
1687 \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
1688 \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
1689 \sloppy\nohyphens{{\ ' a} COPPE/UFRJ como parte dos requisitos
1690 necess{\ ' a}rios para a obten{\c c}{\~ a}o do grau de
1691 \local@degname\ em Ci{\^ e}ncias (\@degree)}
1692 }
1693 \newcommand\foreignabstract@toptext{%
1694 \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
1695 COPPE/UFRJ as a partial fulfillment of the requirements for the
1696 degree of \foreign@degname\ of Science (\@degree)}
1697 }
1698 %
1699 </class>
1700 <*glossary>
1701 actual '='
1702 quote '!'
1703 level '>'
1704 %%% delim_0 " , p. "
1705 delim_0 "\\dotfill "
1706 lethead_flag 0
1707 headings_flag 0
1708 preamble
1709 "\n\\begin{theglossary}\n \\makeatletter"
1710 postamble
1711 "\n \\end{theglossary}\n"
1712 </glossary>
1713 <*class>

```

```

1714 \newcommand{\annex}{
1715     \iflanguage{brazilian}
1716         {\renewcommand{\appendixname}{Anexo}}
1717         {\renewcommand{\appendixname}{Annex}}
1718     \appendix
1719 }
1720 \end{class}

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

1721 \begin{document}
1722 \ProvidesFile{coppe.dbx}[2026/05/23 v0.2 CoppeTeX ABNT datamodel]
1723 % Custom entry types beyond biblatex's standard datamodel (games + TV).
1724 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow}
1725 % Custom fields.
1726 \DeclareDatamodelFields[type=field,datatype=literal]{course}          % "(... em <course>)"
1727 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree}     % degree word
1728 \DeclareDatamodelFields[type=field,datatype=literal]{platform}        % videogame platform
1729 \DeclareDatamodelFields[type=list,datatype=literal]{artist}           % game/illustration art
1730 \DeclareDatamodelFields[type=list,datatype=literal]{director}        % movie/tvshow/video
1731 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
1732 \DeclareDatamodelEntryfields{course,coppedegree}
1733 \DeclareDatamodelEntryfields{game,videogame,boardgame,software,movie,tvshow,video}%
1734 {artist,platform,director,producer,version}
1735 \end{document}
1736 \begin{document}
1737 \ProvidesFile{coppe.bbx}[2026/05/23 v0.1 CoppeTeX ABNT bibliography style]
1738 % ABNT reference list keeps the date at the end (impreta), same for author-date
1739 % and numeric modes, so we derive from the standard style, not authoryear.
1740 \RequireBibliographyStyle{standard}
1741
1742 % Note: standard style never dashes repeated authors, which already matches the
1743 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
1744 \ExecuteBibliographyOptions{%
1745     maxbibnames=99,          % ABNT 4.3.1.3: list all authors by default
1746     giveninits=false,
1747     uniquename=false,
1748     uniquelist=false,
1749     labeldateparts=true, % needed for author-date extradate (1999a/1999b)
1750     sorting=nyt,
1751 }
1752
1753 % Language strings (Disponível em / Acesso em / In, etc.)
1754 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
1755 \DeclareLanguageMapping{english}{english-coppe}
1756
1757 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
1758 \DeclareNameAlias{author}{family-given}
1759 \DeclareNameAlias{editor}{family-given}
1760 \DeclareNameAlias{translator}{family-given}
1761 \DeclareNameAlias{default}{family-given}
1762 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}

```



```

1763 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
1764
1765 % Uppercase the surname (family + prefix) only inside the bibliography list;
1766 % citations keep normal caps (post-2020 rule).
1767 \AtBeginBibliography{%
1768   \singlespacing % entries single-spaced internally (2.4)
1769   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
1770   \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
1771   \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
1772 }
1773
1774 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
1775 %       (no author/editor) -> not bold, first significant word in CAPS (ABNT 4.2). ---
1776 % entry-by-title: plain (not bold). TODO: first significant word in CAPS (ABNT 4.2)
1777 % once a robustly-tested uppercase-first-word macro is in place.
1778 \DeclareFieldFormat{title}{%
1779   \ifboolexpr{ test {\ifnameundef{labelname}} }
1780     {#1}%
1781     {\mkbibbold{#1}}}
1782 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%
1783 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
1784 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
1785 \DeclareFieldFormat[booktitle]{\mkbibbold{#1}}
1786 \DeclareFieldFormat[maintitle]{\mkbibbold{#1}}
1787
1788 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
1789 \DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
1790 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
1791 \DeclareFieldFormat{pages}{p\adddot\addnbspace#1}
1792
1793 % --- Periodical volume / number prefixes ---
1794 \DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
1795 \DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}
1796
1797 % --- "In:" before host title (incollection/inproceedings/inbook) ---
1798 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
1799
1800 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
1801 \DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}
1802
1803 % --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL ---
1804 \DeclareBibliographyDriver{article}{%
1805   \usebibmacro{bibindex}%
1806   \usebibmacro{begentry}%
1807   \usebibmacro{author/translator+others}%
1808   \setunit{\labelnamepunct}\newblock
1809   \usebibmacro{title}%
1810   \newunit\newblock
1811   \printfield{journaltitle}%
1812   \setunit{\addcomma\space}\printlist{location}%
1813   \setunit{\addcomma\space}\printfield{volume}%
1814   \setunit{\addcomma\space}\printfield{number}%
1815   \setunit{\addcomma\space}\printfield{pages}%
1816   \setunit{\addcomma\space}\printdate%

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1817 \newunit\newblock
1818 \printfield{doi}%
1819 \newunit\newblock
1820 \usebibmacro{url+urldate}%
1821 \usebibmacro{finentry}}
1822
1823 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
1824 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
1825
1826 % --- custom entry types reuse existing drivers for now (refine later) ---
1827 \DeclareBibliographyDriver{game}{\usedriver{}{misc}}
1828 \DeclareBibliographyDriver{videogame}{\usedriver{}{misc}}
1829 \DeclareBibliographyDriver{boardgame}{\usedriver{}{misc}}
1830 \DeclareBibliographyDriver{tvshow}{\usedriver{}{misc}}
1831
1832 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
1833 \DeclareSourcemap{%
1834   \maps[datatype=bibtex]{%
1835     % entry-type PT synonyms -> canonical English
1836     \map{step[typesource=livro,typetarget=book]}
1837     \map{step[typesource=artigo,typetarget=article]}
1838     \map{step[typesource=artigodejornal,typetarget=article]}
1839     \map{step[typesource=partedelivro,typetarget=incollection]}
1840     \map{step[typesource=emlivro,typetarget=inbook]}
1841     \map{step[typesource=verbete,typetarget=inreference]}
1842     \map{step[typesource=anais,typetarget=proceedings]}
1843     \map{step[typesource=trabalhodeevento,typetarget=inproceedings]}
1844     \map{step[typesource=periodico,typetarget=periodical]}
1845     \map{step[typesource=relatorio,typetarget=report]}
1846     \map{step[typesource=relatoriotecnico,typetarget=report]}
1847     \map{step[typesource>manual,typetarget>manual]}
1848     \map{step[typesource=correspondencia,typetarget=letter]}
1849     \map{step[typesource=patente,typetarget=patent]}
1850     \map{step[typesource=diverso,typetarget=misc]}
1851     \map{step[typesource=jogo,typetarget=game]}
1852     \map{step[typesource=jogoeletronico,typetarget=videogame]}
1853     \map{step[typesource=jogodetabuleiro,typetarget=boardgame]}
1854     \map{step[typesource=filme,typetarget=movie]}
1855     \map{step[typesource=serietv,typetarget=tvshow]}
1856     \map{step[typesource=programadetelevisao,typetarget=tvshow]}
1857     % thesis family: retype to thesis and preset the (babel-aware) type word
1858     \map{step[typesource=tese,typetarget=thesis]}
1859     \map{step[typesource=dissertacao,typetarget=thesis]}
1860     \map{step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fielddisposition=normal]}
1861     \map{step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fielddisposition=normal]}
1862     \map{step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fielddisposition=normal]}
1863     \map{step[typesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=dscthesis]}
1864     \map{step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tesedoutorado]}
1865     \map{step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phdthesis]}
1866     \map{step[typesource=tesephds,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phdthesis]}
1867     \map{step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
1868     % field PT synonyms (unaccented) -> canonical English
1869     \map{step[fieldsource=titulo,fieldtarget=title]}
1870     \map{step[fieldsource=subtitulo,fieldtarget=subtitle]}

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1871 \map{\step[fieldsource=autor,fieldtarget=author]}
1872 \map{\step[fieldsource=editora,fieldtarget=publisher]}
1873 \map{\step[fieldsource=local,fieldtarget=location]}
1874 \map{\step[fieldsource=ano,fieldtarget=year]}
1875 \map{\step[fieldsource=data,fieldtarget=date]}
1876 \map{\step[fieldsource=edicao,fieldtarget=edition]}
1877 \map{\step[fieldsource=paginas,fieldtarget=pages]}
1878 \map{\step[fieldsource=numero,fieldtarget=number]}
1879 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
1880 \map{\step[fieldsource=organizador,fieldtarget=editor]}
1881 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
1882 \map{\step[fieldsource=tipo,fieldtarget=type]}
1883 \map{\step[fieldsource=nota,fieldtarget=note]}
1884 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
1885 \map{\step[fieldsource=curso,fieldtarget=course]}
1886 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
1887 \map{\step[fieldsource=artista,fieldtarget=artist]}
1888 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
1889 \map{\step[fieldsource=diretor,fieldtarget=director]}
1890 \map{\step[fieldsource=produtor,fieldtarget=producer]}
1891 \map{\step[fieldsource=versao,fieldtarget=version]}
1892 }%
1893 }
1894
1895 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
1896 \setlength{\bibhang}{0pt}
1897 \setlength{\bibitemsep}{\baselineskip}
1898
1899 </bbx>
1900 <*cbx>
1901 \ProvidesFile{coppe.cbx}[2026/05/23 v0.1 CoppeTeX ABNT author-date citations]
1902 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
1903 % author-year cites with normal-caps names (post-2020 rule) out of the box.
1904 \RequireCitationStyle{authoryear-comp}
1905
1906 % --- NBR 10520:2023 author separator -----
1907 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
1908 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
1909 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
1910 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
1911 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
1912 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
1913 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
1914 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
1915
1916 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2.1) -----
1917 % The reference list entry is the CITING work (the one actually consulted).
1918 % \citetapud[<page>]{original}{citing} -> Original (yr) apud Citing (yr[, page])
1919 % \citepapud[<page>]{original}{citing} -> (Original, yr apud Citing, yr[, page])
1920 % #1 = optional postnote (page/loc) on the citing work; #2 = original (not
1921 % accessed); #3 = work consulted. "apud" is a latin expression -> italic (§4.1.2.2).
1922 \DeclareRobustCommand{\citetapud}[3][]{%
1923 \citeauthor{#2}\space(\citeyear{#2})\space\mkbibemph{apud}\space\textcite{#1}{#3}}
1924 \DeclareRobustCommand{\citepapud}[3][]{%

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1925 \mkbibparens{%
1926   \citeauthor{#2}\addcomma\space\citeyear{#2}\space
1927   \mkbibemph{apud}\space
1928   \citeauthor{#3}\addcomma\space\citeyear{#3}%
1929   \ifblank{#1}{ }\{\addcomma\space\mkpageprefix[pagination]{#1}}\}}
1930 \</cbx>
1931 \<*lbxbr>
1932 \ProvidesFile{brazilian-coppe.lbx}[2026/05/23 v0.1 CoppeTeX brazilian strings]
1933 \InheritBibliographyExtras{brazilian}
1934 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc}
1935 \DeclareBibliographyStrings{%
1936   inherit      = {brazilian},
1937   availablefrom = {{Dispon\'\i vel em}{Dispon\'\i vel em}},
1938   urlseen      = {{Acesso em}{Acesso em}},
1939   in           = {{In}{In}},
1940   mscdiss      = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
1941   dscthesi     = {{Tese de Doutorado}{Tese de Doutorado}},
1942   phdthesis    = {{Tese de Doutorado}{Tese de Doutorado}},
1943   tcc          = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
1944 }
1945 \</lbxbr>
1946 \<*lbxen>
1947 \ProvidesFile{english-coppe.lbx}[2026/05/23 v0.1 CoppeTeX english strings]
1948 \InheritBibliographyExtras{english}
1949 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc}
1950 \DeclareBibliographyStrings{%
1951   inherit      = {english},
1952   availablefrom = {{Available from}{Available from}},
1953   urlseen      = {{Access on}{Access on}},
1954   in           = {{In}{In}},
1955   mscdiss      = {{M\adddot Sc\adddot Dissertation}{M\adddot Sc\adddot Diss\adddot}},
1956   dscthesi     = {{D\adddot Sc\adddot Thesis}{D\adddot Sc\adddot Thesis}},
1957   phdthesis    = {{Ph\adddot D\adddot Thesis}{Ph\adddot D\adddot Thesis}},
1958   tcc          = {{Undergraduate Thesis}{Undergraduate Thesis}},
1959 }
1960 \</lbxen>
1961 \<*numbbx>
1962 \ProvidesFile{coppe-numeric.bbx}[2026/05/24 v0.1 CoppeTeX ABNT numeric bibliography]
1963 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
1964 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
1965 % the PT->EN source map and the language mappings) and only adds the [n] list
1966 % label and the numeric sorting on top.
1967 \RequireBibliographyStyle{coppe}
1968
1969 % ABNT numeric system lists references in the order they are first cited
1970 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
1971 % numbered list. 'labelnumber' generates the [n] field.
1972 \ExecuteBibliographyOptions{labelnumber,sorting=none}
1973
1974 % Bracketed labels, both in the list and for any shorthand entries.
1975 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
1976 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
1977
1978 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),

```

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1979 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
1980 \defbibenvironment{bibliography}
1981   {\list
1982     {\printtext[labelnumberwidth]{%
1983       \printfield{labelprefix}%
1984       \printfield{labelnumber}}}
1985     {\setlength{\labelwidth}{\labelnumberwidth}%
1986       \setlength{\leftmargin}{\labelwidth}%
1987       \setlength{\labelsep}{\biblabelsep}%
1988       \addtolength{\leftmargin}{\labelsep}%
1989       \setlength{\itemsep}{\bibitemsep}%
1990       \setlength{\parsep}{\bibparsep}}%
1991     \renewcommand*{\makelabel}[1]{\hss##1}}
1992   {\endlist}
1993   {\item}
1994
1995 \

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Acknowledgments

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